

Academic Qualifications			
Year	Degree/Certificate	Institute	CPI/%
2023–Present	B.Tech	Indian Institute of Technology Kanpur	7.1/10
2022	CBSE (XII)	SKP Vidya Vihar, Bhagalpur	95.4%
2020	ICSE (X)	St. Teresa’s School, Bhagalpur	96.4%
Work Experience			
IoT Security Auditing Framework Internship, C3iHub, IIT Kanpur Rahul Shukla			Sep’26–Oct’26
• Designing and setting up a secure IoT Security Testbed to emulate real-world distributed and connected IoT systems and networks. • Performing hardware security assessment through interface identification, UART/SPI Flash testing, firmware extraction, and bootloaders. • Developing automated reconnaissance and scanning for Wi-Fi networks, covering discovery, device identification, and security assessment. • Performing Wi-Fi security testing, including deauthentication attacks and Aircrack-ng-based analysis in controlled environments. • Developing an integrated framework for device scanning, hardware interface analysis, firmware extraction, and Wi-Fi security auditing.			
Key Projects			
Pipelined AES-128 Hardware Implementation Course Project Prof. Urbi Chatterjee — CS666			Sep’25–Nov’25
• Designed and implemented a high-throughput AES-128 encryption core using an 11-stage outer-round pipelined architecture on FPGA. • Enhanced overall performance by parallelizing key expansion with cipher rounds, removing critical timing delays and improving throughput. • Functionally verified the design using standard NIST test vectors through detailed simulation and waveform-level timing analysis results. • Validated on PYNQ-Z2 (Zynq-7000 SoC) via AXI4-Lite, delivering complete RTL source code, testbenches, and Vivado project files.			
Fusion Power Prediction using PINNs Course Project Prof. Salman Khan — CHE657			Sep’25–Nov’25
• Developed and compared machine learning models, including a Physics-Informed Neural Network and standard neural networks. • Designed a custom physics-based loss function to enforce steady-state fusion power laws for physically meaningful, consistent predictions. • Preprocessed and normalized 10,000+ experimental runs, using scaling to manage extreme values and ensure numerical stability. • Evaluated model performance using error-based metrics, highlighting the PINN’s advantage in efficient, physically grounded prediction.			
Gene Expression Prediction from Histone Modification Signals Course Project Prof. Sana Khanum — CHE629			Feb’26–Apr’26
• Predicted binary gene expression (ON/OFF) from 5 histone modification marks across 100 spatial bins spanning $\pm 5,000$ bp around the transcription start site, on 19,799 genes of the REMC E003 human embryonic stem cell dataset. • Designed and benchmarked a CNN against a hierarchical BiLSTM with dual attention, using bin-level spatial attention and mark-level combinatorial attention to model both long-range spatial dependencies and inter-mark correlations. • Achieved 0.788 test AUC with the attention model — matching the CNN using 11× fewer parameters (59K vs 644K) and 5× higher minority-class F1 — closing in on the published DeepChrome and AttentiveChrome baselines. • Countered severe class imbalance (79.2% inactive genes) with Focal Loss, LR scheduling on validation AUC, gradient clipping and early stopping; showed learned attention weights independently re-discover known epigenetic logic (TSS focus, H3K4me3 for active, H3K27me3 for repressed) without manual annotation.			
2D Poisson Solver: GPU Acceleration Course Project Prof. Mahendra Kumar Verma — IDC606			Jan’26–Apr’26
• Engineered a high-performance spectral 2D Poisson equation solver on an NVIDIA RTX PRO 6000 GPU using CuPy and NVIDIA Warp. • Achieved an 8,000× runtime speedup (28.9s to 0.36ms for a 1024×1024 grid) using a GPU-accelerated Fast Fourier Transform (FFT). • Replaced the vectorized Red-Black SOR baseline with an $O(N^2 \log N)$ spectral FFT algorithm, eliminating iterative convergence bottlenecks. • Integrated DLPack zero-copy interoperability for a complete on-GPU data path, achieving machine-precision accuracy ($L_2 \sim 10^{-16}$).			
Machine Learning Foundations Mini-Projects Prof. Rajesh M. Hegde — EE708			
• Completed a semester-long series of weekly assignment mini-projects covering the full data-science pipeline — statistical inference, regression, clustering, Gaussian Mixture Models, decision trees, and deep learning (CNNs, RNNs, autoencoders). • Implemented core algorithms from scratch in NumPy, including a perceptron trained by gradient descent with decision-boundary visualisation and a feedforward neural network trained via SGD; derived backpropagation weight updates for ReLU MLPs by hand. • Applied Pandas, Matplotlib and scikit-learn to real datasets for feature engineering, normalisation and binning, least-squares fitting with residual diagnostics, hierarchical clustering with dendrograms, vector quantisation, and Gini/SSE-based tree induction.			
ML Kickstart Association for Computing Activities, IIT Kanpur			May’25–Jun’25
• Built a machine learning model to classify songs by genre using the Spotify Songs and their Characteristics dataset. • Implemented ML algorithms including linear regression, classification models, and neural networks using scikit-learn and TensorFlow. • Performed exploratory data analysis (EDA), feature engineering, and thorough data preprocessing on real-world datasets. • Gained foundational experience in model training, validation, and interpreting results for regression and classification tasks.			
Mesoscale and Continuum Simulation Course Project Prof. Indranil Saha Dalal — CHE643			Jun’26–Jul’26
• Built NumPy simulators from scratch; benchmarked explicit Euler on Stokes settling showing accuracy collapse as $\Delta t \rightarrow \tau$ with large timesteps. • Extracted diffusivity from multiple-time-origin MSD fits ($D = \text{slope}/6$) on 10^5-step 3D Langevin trajectories for reliable transport estimates. • Handled periodic boundaries by tracking wrapped and unwrapped coordinates in parallel, keeping box crossings out of the MSD calculation. • Ran a 10^7-step FENE dumbbell simulation ($\nu = 500$): RMS end-to-end 22.74 vs. theory $\sqrt{\nu} = \mathbf{22.36}$, showing strong agreement. • Mapped shear-thinning across 8 shear rates spanning 3.5 decades via the Kramers stress tensor under steady shear flow. • Implemented DPD (81 beads, $\rho = 3$) with fluctuation–dissipation-coupled noise; MSD resolved ballistic-to-diffusive crossover across timescales. • Grafted a 5-bead chain into 76 solvent beads: RMS end-to-end 2.494 vs. Spenley’s 2.525 (1.2% error), closely matching theory.			
MediQ Mentor, Society of Civil Engineers, IIT Kanpur			Dec’25–Jan’26
• Mentored ~25 students through end-to-end ML workflow, guiding them from data preprocessing to a deployed disease-diagnosis web application. • Guided design of a multi-class disease classification pipeline using a Random Forest ensemble (300 estimators) with balanced class weighting			

- and `sqrt` feature selection for symptom-based diagnosis.
- Directed the ML pipeline build, covering **symptom feature normalization**, **duplicate-column handling**, **label encoding**, **stratified 80:20 train-test splitting**, and automated model evaluation using accuracy and classification reports.
- Oversaw implementation of a **probabilistic inference engine** using `predict_proba()` to rank and visualize **Top-5 differential diagnoses**, with confidence scores and symptom-severity weighted risk scoring.
- Supervised development of a **Flask REST API** for real-time inference, integrated with a structured knowledge base of disease descriptions, precautions, diets, medications, and lifestyle recommendations.

GIFSleuth: GIF Forensics & Steganography Scanner | Self Project |

Apr'25

- Built a **zero-dependency Python** GIF security scanner that parses **GIF87a/89a** byte streams and runs **7 detectors** on non-rendered regions.
- Detected **GIFAR**, trailer-appended payloads, content-sniffing **XSS polyglots**, embedded web shells, and **gifshuffle steganography**.
- Combined **19 embedded file signatures** with **Shannon entropy** analysis to detect encrypted or packed payloads and report offsets for verification.
- Reduced false positives by scanning renderer-ignored carrier regions and stripping power-of-two palette padding from LZW image data.
- Designed a **self-registering plugin architecture** with batch scanning, **JSON/CSV/ANSI** reporting, and contractual exit codes for CI pipelines.
- Wrote **143 stdlib unittest** cases with byte-level GIF fixtures and a **GitHub Actions** pipeline across **Python 3.11–3.13**.

Algorithmic Problem Solving with Advanced Graph Techniques | Self Project |

May'25–Jun'25

- Implemented graph algorithms including BFS/DFS, shortest paths using Dijkstra, Bellman-Ford, Floyd-Warshall, and minimum spanning trees.
- Applied **Kosaraju's and Tarjan's algorithms** for strongly connected component detection in complex large-scale networks.
- Developed and optimized solvers for Sudoku, maze navigation, and Rubik's Cube using **A* search** and other pathfinding strategies.
- Evaluated and compared algorithm performance through detailed complexity analysis and extensive hands-on benchmarking experiments.

Integrated Single-Train Ammonia Plant | Course Project | Prof. Harshvardhan Katkar — CHE251

Oct'24–Nov'24

- Designed a **1500 t/d integrated single-train ammonia plant** flowsheet covering reforming, CO₂ removal, methanation, and synthesis.
- Modeled the process in **Aspen Plus**, computing reaction extents for primary/secondary reforming, shift, and Haber-Bosch synthesis.
- Performed detailed mole and component balances across all **9 process streams**, validating simulation outputs against given plant data.
- Analyzed hydrogen recycle, purge control, and heat integration strategies to improve ammonia yield and overall energy efficiency.

Fluroxypyr Production EHS Analysis | Course Project | CHE261 |

Feb'25–Mar'25

- Conducted an **Environmental, Health and Safety** analysis for Fluroxypyr production, focusing on hazardous waste streams and safety risks.
- Designed a complete **Zero Liquid Discharge (ZLD)** treatment system incorporating catalyst recovery, evaporation, and crystallization stages.
- Calculated sustainability metrics: **Atom Economy (61.58%)**, **E-Factor (0.624)**, and **EQ-Factor (1.56)** to evaluate green process efficiency.
- Studied and documented exposure limits and disposal regulations for chemicals including **Toluene, NaOH, Ethanol, and TBAI**.
- Co-authored the final technical report including mass balance, regulatory compliance, and safety measures for lab and industrial scale-up.

LSTM Neural Network for Time Series Prediction | Self Project |

May'26–July'26

- Studied and reproduced a **stacked LSTM architecture in Keras/TensorFlow** for univariate and multivariate time-series forecasting, training on both synthetic sine-wave signals and historical stock-market data across varying forecast horizons.
- Implemented and compared three forecasting strategies—**point-by-point prediction**, **full-sequence prediction**, and **multi-sequence (windowed) prediction**—to systematically evaluate trade-offs between short-term accuracy and long-horizon forecast stability under different settings.
- Built the data pipeline with **sliding-window sequence generation** and **per-window normalization** to handle non-stationary financial price data and prepare multi-dimensional input tensors for robust network training and evaluation.
- Investigated **recursive multi-step forecasting**, identifying and visualizing **prediction drift** as forecast errors accumulated progressively over longer prediction horizons compared against corresponding ground-truth sequences.
- Tuned **network depth, dropout regularization, and sequence length** hyperparameters, and benchmarked training/inference performance across multiple configurations to understand their effect on convergence, generalization, and forecast quality.

TECHNICAL SKILLS

Programming Languages	Frameworks & Libraries	Tools & Softwares
C, C++, Python, Bash, MATLAB, SQL, HTML, CSS, Verilog	NumPy, Pandas, TensorFlow, OpenCV, Matplotlib, scikit-learn	Git, Linux, Jupyter Notebooks, Vim, Xilinx Vivado, $\LaTeX$, Aspen Plus, Raspberry Pi, Arduino

RELEVANT COURSES

(*: Ongoing)

Intelligent Pattern Recognition*	Fundamentals of Data Science and Machine Intelligence	Computer Methods for Engineers
Fundamentals of Computing-I	Machine Learning in Chemical Engineering	Introduction to Electronics
Bayesian Models and Data Analysis	Hardware Security for Internet-of-Things	Multivariable Calculus
Computer Arithmetic on FPGA*	Mesoscale and Continuum Simulations In Chemical Engineering	Data Structure & Algorithm
AI in Systems Biology	High Performance Computing with Applications	Linear Algebra & ODE

POSITIONS OF RESPONSIBILITY

Senior Executive | Udghosh'24, IIT Kanpur

Oct'24

- Managed logistics and coordination for **100+ football participants** from multiple colleges, ensuring seamless event operations.
- Oversaw team registrations, match scheduling, and **ground-level execution** to ensure smooth conduct of the tournament.
- Acted as the primary **point of contact** between the organizing committee and team captains for football events and related activities.

EXTRA-CURRICULAR ACTIVITIES

- Mentored **20 students** on the MediQ project, leading hands-on sessions in machine learning and web development for an ML web application.
- Served as a cadet in the **National Cadet Corps (NCC)** under the **2 UP Composite Technical Regiment (2UPCTR)**.
- Volunteered at **Antaragni'23**, assisting in the planning, organization, and execution of events during IIT Kanpur's annual cultural fest.
- Built and deployed a static **portfolio website** on GitHub Pages and learned responsive design principles and deployment workflows.
- Passionately pursue music as a hobby by recording and uploading vocal covers of Bollywood songs on a personal **YouTube channel**.